

LI Leyan | 李乐岩

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GitHub: <https://github.com/lionlepower>

Education

University of Edinburgh

MSc High Performance Computing with Data Science

Sep 2025 – Present (Expected Sep 2026)

Relevant Topics:

Parallel Computing, MPI Programming, Performance Engineering, Distributed Systems

University of Liverpool

BSc Computer Science (First Class Honours)

Sep 2023 – Jun 2025

Xi'an Jiaotong-Liverpool University

BSc Information and Computer Science

Sep 2021 – Jun 2023

Summary

High Performance Computing and Data Science MSc student with strong experience in parallel and distributed computing. Skilled in C/C++, Python, MPI, and OpenMP, with practical experience running large-scale experiments on the ARCHER2 supercomputing system. Interested in scalable systems, performance engineering, and the intersection of HPC and machine learning.

Technical Skills

Programming Languages

C/C++, Python, SQL

Parallel / HPC

MPI, OpenMP, PETSc, Slurm, ARCHER2

Machine Learning

PyTorch, ePyMARL

Tools & Environment

Linux, Git, VS Code

Projects

1. PETSc Benchmark and Performance Analysis Suite

GitHub: <https://github.com/lionlepower/petsc-benchmark-suite>

Developed a reproducible benchmarking framework for analysing PETSc linear solver performance under different parallel configurations.

Designed automated Slurm job pipelines to conduct strong scaling and hybrid MPI+OpenMP experiments on the ARCHER2 supercomputer.

Built Python-based analysis tools to process runtime logs, generate CSV datasets, and produce speedup and parallel efficiency visualisations.

Performed performance analysis comparing MPI-only and hybrid execution models, identifying communication overhead and scalability limits.

2. HPC Mini Applications (MPI / OpenMP) — in progress

Implemented a collection of classical high-performance computing mini-applications in C, including:

- 2D heat diffusion simulation using stencil computation
- parallel matrix multiplication
- N-body simulation

Designed MPI-based domain decomposition and halo exchange mechanisms for distributed-memory parallelism.

Integrated OpenMP thread-level parallelism and evaluated hybrid MPI+OpenMP configurations.

Conducted scaling experiments to analyse parallel efficiency, load balance, and memory behaviour under different process and thread configurations.

Value Function Factorisation in Multi-Agent Actor-Critic Methods

Investigated factorised critic architectures (VDN, QMIX) within Multi-Agent Proximal Policy Optimization (MAPPO) to improve coordination in cooperative multi-agent environments.

Implemented custom MARL algorithms using Python and PyTorch based on the PyMARL / ePyMARL framework.

Conducted experimental evaluation in Matrix Game and Predator–Prey environments, analysing convergence behaviour and coordination performance.

Explored improvements to reward shaping and critic factorisation strategies to enhance learning stability.

Research Experience

String Art Generator Based on Radon Transform

Summer Undergraduate Research Programme

Jun 2024 – Aug 2024

Developed an algorithmic pipeline that converts images into string-based artwork using Radon transform techniques.

Implemented image preprocessing and contrast enhancement algorithms to improve feature extraction.

Designed a parameter control interface enabling interactive adjustment of artistic generation parameters.

Publication

Li, Leyan (2024)

Convolutional Neural Networks (CNNs)-based Medical Image Analysis

International Conference on Engineering Management, Information Technology and Intelligence (EMITI 2024)

Internship Experience

Anhui Esou Information Technology Co., Ltd

Front-end Developer Intern

Aug 2023 – Sep 2023

Developed a parking management interface using HTML, CSS, and JavaScript.

Implemented dynamic UI components including real-time vehicle count monitoring and weather information display.

Collaborated with backend engineers to integrate front-end components with REST APIs.